RUDRAKSH SYAL

Data Science Lead | Forward Deployed Agentic AI & LLM Systems

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PROFESSIONAL PROFILE

Hands-on data science lead with 11+ years in Python, data science, and ML, turning ambiguous business problems into shipped ML and LLM systems with the people who use them, from business users and clients to executive stakeholders up to the COO and CPO. Earned adoption of an LLM tool-calling pricing layer (Claude on AWS Bedrock) from pricing teams across 7 U.S. regions, moving from a parallel run to go-live; 85–90% of its recommendations are accepted. Now building agentic AI workflows from scratch for two client engagements.

TECHNICAL SKILLS

Agentic AI & LLM Delivery: LLM tool/function calling, Anthropic Claude (Sonnet, Haiku, Opus) via AWS Bedrock, OpenAI Agents SDK, LangChain/LangGraph, MCP, RAG, hybrid search, Llama 3.1 8B, LLM extraction

Evaluation & Human-in-the-Loop: Parallel runs, approve/reject feedback loops, LLM-as-a-judge, golden sets, Precision@K/Recall@K, guardrails, LangSmith tracing, shadow mode, backtests, pre/post measurement

Integration & Data: AWS (Bedrock Knowledge Bases, SageMaker, OpenSearch Serverless), Snowflake, Salesforce, Databricks, Azure, MLflow, Airflow, Python, SQL, R, Spark, Flask

Modeling: XGBoost, Random Forest, forecasting, optimization, causal inference (BSTS, synthetic control)

PROFESSIONAL EXPERIENCE

Data Science Lead | Hana E Solutions **Oct 2026–Present**

Remote, San Jose, CA

- **Client agentic AI delivery:** Founding data science contributor building agentic AI workflows from scratch for two clients, mainly in pricing. Both are at the proof-of-concept stage.

Senior Data Scientist | Vista **Oct 2021–Jul 2026**

San Jose, CA (Remote)

Agentic AI Shipped with Business Users

- **Earned adoption with pricing managers:** Over a six-quarter program, worked with fleet pricing managers (5–8 per region) and revenue leaders (2 per region) across 7 U.S. regions, reporting up to the COO and CPO, to adopt an agentic pricing layer. A Q6 parallel pricing run won their buy-in, and the layer went live in Q7. It now handles ~240 pricing requests a day across 3 fleets, and rejections fell from ~30% to 10–15%.
- **Built for human approval:** Built the layer in custom Python with LLM tool calling on Claude Sonnet, Haiku, and Opus via Bedrock, on top of the Dynamic Pricing Engine. The LLM calls forecasting and Snowflake tools and explains pricing drivers from structured results. Pricing managers approve or reject every recommendation.
- **Feedback loop & guardrails:** Paired LLM-as-a-judge evaluations with approve/reject outcomes, using acceptance rate as the success metric. Wrote guardrails in custom Python and traced runs with LangSmith.
- **Contract RAG & LLM extraction:** Built hybrid retrieval over 4,000+ contracts (Bedrock Knowledge Bases, OpenSearch Serverless, SageMaker-hosted Llama 3.1 8B), evaluated on golden question–clause sets (Precision@K/Recall@K). With product and engineering, turned refund terms for ~600 operators into cancellation-policy badges; 84.7% were badge-ready automatically.

End-to-End Decision Systems

- **Pricing engine, problem to production:** Led two data scientists to replace static pricing bands with a supply-demand equilibrium engine across seven U.S. regions. Used interpretable, bounded components so users would trust the output. One 43-aircraft fleet saw 28% fewer off-fleet bookings in month one.
- **Safe relaunch via shadow mode:** Built a gradient-boosted EU cost model, validated against shadow-mode invoices and backtests before relaunch. Margin went from −8% to +17%, and loss-making bookings fell from 55% to 20%.
- **Integrations into business systems:** Built XGBoost lead-propensity scores integrated into Salesforce, and raised booking forecast accuracy for 14 fleet categories from 71% to 78%.

Senior Data Scientist | 84.51° (Kroger) **Jul 2019–Oct 2021**

Cincinnati, OH

- **Client-facing measurement for CPG brands:** Represented the data science team directly with CPG clients; presented BSTS and synthetic-control results for Quaker endcaps (4% lift vs. a 10% benchmark), which informed the decision to pause an Eastern-division rollout.
- **Optimization to pilot:** Optimized bakery footage with Random Forest and integer programming. Nine pilot stores recorded 14% higher early sales (directional).

Data Scientist | Practo Technologies **Jun 2015–Apr 2018**

Bangalore, India

- **Internal tools that replaced a vendor:** Built R/Shiny and Flask reporting apps that replaced a $4K/month-per-person analytics platform.

PROJECTS

- **Dynamic Pricing Engine** (personal rebuild on anonymised data)
  github.com/rudrakshsyal/dynamic_pricing_engine
  An OpenAI Agents SDK layer writes tool-bound revenue briefs, checked for citations and numbers. A budget-capped GPT-5.5 research agent was benchmarked against production-style ML forecasts; results were mixed, so the ML forecasts were kept.
- **Sidekick Agent** (personal project)
  github.com/rudrakshsyal/the_sidekick_agent
  A LangChain/LangGraph agent with MCP tools (Playwright browser, sandboxed filesystem), an evaluator loop with up to 3 retries, PII and model-call-limit guardrails, and human-in-the-loop approval.

LEADERSHIP, PATENT & EDUCATION

Leadership: Led two data scientists on pricing systems; started cross-functional learning sessions for 60+ colleagues.

Patent: U.S. Patent No. 12,314,757 B1 (2025): Efficient resource scheduling using adaptive scheduling criteria (co-inventor).

Purdue University — M.S., Business Analytics & Data Science (2019)

BITS Pilani — B.E. (Hons.) and M.Sc., Applied Mathematics (2015)